



Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

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Project Title

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1. Problem Statement

Ground Glass Opacities (GGOs) are faint cloudy areas seen in lung CT scans. They can be early signs of diseases such as pneumonia, lung infections, or even cancer. Finding these areas by hand is difficult and takes a lot of time because GGOs have low contrast and spread out gradually in the lungs. Different doctors may also interpret them differently, which can affect diagnosis accuracy.

Although deep learning methods are used in medical imaging, many existing systems mainly focus on identifying whether a disease is present, rather than clearly showing where GGOs are located in the CT scan.

This project proposes an image processing and computer vision based tool that automatically highlights GGO regions in lung CT scans. The system processes raw CT images and converts them into clear and easy to understand visual outputs. This tool is designed to support radiologists by improving the accuracy of interpretation and helping with early detection and monitoring of lung abnormalities.

2. Objectives

- Improve the quality of the chest CT scans by removing the noise, normalizing the image intensity values, and converting the data into a format suitable for processing.
- Segmentation of the lung regions from the CT volumes using basic intensity based methods, and morphological image processing techniques
- Show the detected GGO areas on the CT images by overlaying them visually or using pseudo coloring enhancement to highlight their visibility.
- Test the efficiency of the proposed image processing workflow on a publicly available chest CT dataset.
- Evaluate the effectiveness of the proposed image processing pipeline using a publicly available chest CT dataset.
- Design and show a simple prototype application which focuses on clear image visualization and the ease of interpretation of the results.

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3. Literature Review

Computed tomography is a cornerstone of pulmonary diagnostics. It reveals subtle parenchymal changes, especially ground-glass opacities. Ground-glass opacities are faintly hazy areas indicating inflammation or early neoplastic processes. Manual identification of ground-glass opacities is the reference standard. Manual identification is time-intensive and has inter-observer variability. This review surveys major approaches: preprocessing, segmentation, detection, and vessel suppression. It highlights limitations that suggest a need for interactive visualization tools.

3.1. Study 1: Hounsfield Unit Based Lung Segmentation

Early automated lung segmentation used intensity thresholding. Air-filled lung tissue has different radiodensity than surrounding structures. Hounsfield unit ranges provide reliable boundaries for tissue types [1]. Simple binary separation distinguishes lung parenchyma from other thoracic components. A 3D approach combined thresholding, dynamic programming, and morphological operations [2]. Dynamic programming helps in regions with faint intensity gradients. Morphological steps remove artifacts like disconnected fragments and incomplete contours. These techniques set early performance standards for segmentation. Accessible implementation recipes followed. One workflow used Hounsfield thresholds of -1000 to -500 HU and morphological closing [3]. Processing lungs independently preserves anatomy in the hilar region. Such pipelines use Python libraries for experimentation and adaptation.

3.2. Study 2: Advanced Segmentation for Diseased Lungs

Disease alters tissue attenuation and texture, challenging thresholding. Interstitial lung disease segmentation combined thresholding and texture maps [4]. Texture maps use gray-level co-occurrence matrices for spatial arrangement. This method reached 82% overlap with expert outlines. Hybrid strategies integrate fuzzy clustering and morphological refinement. Fuzzy c-means clustering is followed by hole filling, dilation, and reconstruction [5]. Attention is given to juxta-pleural nodules blending into the chest wall. Fuzzy clustering handles gradual density transitions better than crisp thresholds. Morphological reconstruction closes gaps without distorting boundaries. These steps help when consolidation or fibrosis obscures lung margins.

3.3. Study 3: Ground Glass Opacity Detection Algorithms

After segmentation, the task is to separate ground-glass density from normal tissue. Markov random field models use spatial context among voxels [6]. They achieved 86.94% sensitivity and 94.33% specificity. Morphological vessel removal reduces false positives. Other groups use statistical characterization of candidate regions. First-order statistics describe intensity distributions for classification [7]. Entropy is especially discriminative for ground-glass texture. Texture-oriented classifiers work well for nodular ground-glass

opacities. Boosted k-nearest-neighbor and density estimation create texture likelihood maps [8]. This method detected all test nodules with few false positives. COVID-19 increased demand for computationally lean methods. Attention-thresholding improved Dice scores by 8.9% with low computational cost [9]. It focuses computation on diagnostically salient regions. Volumetric geometric analysis enables new characterization. PointNet++ achieved 98% accuracy and 92% IoU for 3D ground-glass regions [10]. Minkowski tensors provide shape descriptors for ground-glass opacities. Geometric features allow finer subtyping and may correlate with disease mechanisms.

3.4. Study 4: Pulmonary Vessel Segmentation and Suppression

Pulmonary vessels confound ground-glass detection due to similar attenuation. Multiscale Hessian-based filtering enhances vessels. A 3D multiscale framework segmented the vascular tree with 96.2–97.0% accuracy [11]. Eigenvalue signatures of tubular geometry are used across scales. Dynamic contour evolution applies geometric constraints. Morphological processing and elliptical fitting use vessel aspect ratios [12]. Edge tracking maintains continuity along vessel paths. Tubular morphology helps reject non-vascular detections. Comprehensive pipelines combine thresholding, closing, and Hessian filters. Distance-transform processing quantifies vessel proximity [13]. These tools support region-aware vessel suppression.

3.5. Gap Analysis

Important limitations remain despite progress. Most GGO detection systems are fully automatic with little user intervention. Radiologists cannot fine-tune parameters for individual anatomy or disease. This limits clinical integration. Vessel suppression uses globally uniform settings. Vascular density varies from hilar region to lung periphery. Uniform parameters cause under-suppression centrally or tissue loss peripherally. Validation studies focus on overall metrics, not failure modes. Clinicians lack guidance on where improvements are needed. Few methods offer volumetric quantification with slice-by-slice navigation. Direct correlation between 3D extent and clinical severity is limited. Transparency control for overlay visualization is rarely addressed. Efficiency evaluations focus on algorithmic steps, not end-to-end workflow. Total processing time determines clinical realism. These limitations motivate interactive visualization tools. Tools should provide adjustable parameters and regionally adaptive processing. Detailed failure-mode analysis and volumetric navigation are needed. Overlay transparency control and workflow optimization are important. Such capabilities would bridge research automation and clinical utility.

4. Methodology

(Minimum word limit: 1000 words) Describe the approach, tools, and algorithms you will use. Mention the main image processing or computer vision techniques (e.g., segmentation, feature extraction, classification) and the software frameworks (e.g., OpenCV, MATLAB, TensorFlow).

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5. Dataset / Data Collection

(Maximum word limit: 100 words) Provide details of the data its source, number of images, format, and characteristics. If you plan to collect your own data, explain how you will ensure data quality, diversity, and ethical use.

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6. Expected Outcomes

(Maximum word limit: 100 words) Summarize the expected outputs (eq: demo, code files, presentation) and performance metrics (e.g., accuracy, clarity, speed). Explain how your results will demonstrate the effectiveness of your image processing or computer vision approach. Write the outcomes in point form.

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7. Percentage of Image Processing & Computer Vision Involvement

(Approximately 50–100 words) State clearly how your project satisfies the requirement of at least 65% image processing and computer vision components. For example, describe which parts involve preprocessing, segmentation, or model development etc. And if you are planning to use any tools may be you can mention.

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A minimum of ten (10) references must be included in the References section.